

Parniyan Malekzadeh

Computer Engineering Graduate

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SUMMARY

Computer Engineering (Secure Computing) B.Sc. graduate, GPA 18.05/20.0, top 0.3% in the National University Entrance Exam. Head TA in cryptography and secure systems. Research focus: where a budgeted defender should look, how cryptographic implementations leak, and whether a flow classifier can be evaded. Implemented the sampling-budget IDS game, simulated SPA of textbook RSA-CRT, and constrained adversarial examples against a flow NIDS.

EDUCATION

Isfahan University of Technology

2021 – 2026

B.Sc. in Computer Engineering (Secure Computing)

Isfahan, Iran

- GPA: 18.05/20.0; Major GPA: 4.0/4.0. Top 10% of Computer Engineering students.
- B.Sc. project (course requirement, off the research thread): platform connecting advertisers and promoters. Supervisor: Dr. Hossein Falsafain.

National Organization for Development of Exceptional Talents (Sampad)

2018 – 2021

High School Diploma in Mathematics and Physics

Isfahan, Iran

- GPA: 19.8/20.0

RESEARCH INTERESTS

- Network security: budgeted IDS sampling, SDN, intrusion detection
- Applied cryptography and side-channel analysis (SPA, RSA-CRT)
- Adversarial machine learning for security

EXPERIENCE

Kasra Engineering Co.

Aug. 2024 – Sept. 2024

Software Engineering Intern

Isfahan, Iran

- Developed enterprise Android modules in Kotlin (MVVM); leak hunting with Android Profiler and LeakCanary.
- REST with Retrofit, local data with Room; tests with JUnit, Espresso, and Mockito.

Isfahan University of Technology

2021 – 2025

Teaching Assistant (Head TA: Secure Computing, Secure Software Development)

Isfahan, Iran

- Secure Computing (Head TA): coordinated the TA team; assessments on protocols, secure design, threat modeling.
- Secure Software Development (Head TA): designed and graded curriculum projects.
- Fundamentals of Cryptography: RSA, simple power analysis (SPA), Chinese Remainder Theorem.
- Also TA: Automata Theory, Artificial Intelligence, IT Fundamentals (Zero Trust, Packet Tracer), Database Design, Linear Algebra.

PROJECTS

Game-theoretic IDS sampling (research)

2026

github.com/parmalvar/IDS-Sampling-Game

- Implemented the sampling-budget IDS game (min-cut / max-flow allocation) and compared it to uniform and random sampling on multiple topologies.
- Under a tight budget ($Bs/MF \approx 0.30$) on a 3×3 grid with decoy links, min-cut reached detection 0.300 vs 0.161 uniform (87% relative gain).

Simulated SPA on textbook RSA-CRT (research)

2026

github.com/parmalvar/RSA-CRT-SPA

- Recovered d_p from a single simulated square-and-multiply trace and factored 1024-bit n via gcd on 100% of seeded trials.
- Dummy multiply and a Montgomery ladder defeated that distinguisher (0% exact d_p); message blinding still leaked d_p .

Constrained adversarial examples against a flow NIDS (research)

2026

github.com/parmalvar/Adversarial-Flow-NIDS

- Unconstrained PGD reached 82% ASR at scaled L_{∞} $\epsilon=0.5$ on a CIC-IDS2017 DoS flow MLP; a padding/delay mutability mask lowered ASR to 46%.
- Constrained examples still transferred to a random forest at 61%; PGD adversarial training cut constrained ASR to 8%.

Game-Theoretic Network IDS (course derivation)

2025

github.com/parmalvar/Game-Theoretic-Network-IDS

- Course report: zero-sum sampling game, minimax, max-flow/min-cut allocation, cooperating attackers. Prior work for the evaluation repo above.

Network Security CTF & penetration testing

2025

github.com/parmalvar/NetSec-CTF-Writeup

- Kill-chain on a lab network (Zimbra XXE, PHP type juggling, Wiener RSA); documented as a writeup, not a product.

Mininet / Open vSwitch labs

2023

github.com/parmalvar/Mininet-OVS-Projects

- SDN-style topologies, Wireshark on DNS/HTTP/TCP; supporting systems background for the sampling-game Mininet sketch.

Other coursework repos (not lead research): CERT incident-response literature review; Deep RL reproducibility homework; Netflix recommender; video-platform database; Mitnick lab.

HONORS AND AWARDS

Isfahan University of Technology

2025

Ranked among the top graduating Computer Engineering students

- Direct M.Sc. admission for outstanding B.Sc. performance; Brilliant Talents Foundation / Exceptional Students.

National University Entrance Exam (Konkur)

2021

Ranked 591 among more than 160,000 participants (top 0.3%), Mathematics and Physics

SELECTED COURSEWORK

Secure Software Development 20.00 · Cryptography 20.00 · Network Security Lab 20.00 · Secure Computing 18.20 · Computer Networks 18.60 · Game Theory 18.30 · Software Testing 18.43 · Databases I 19.06 · Computer Architecture 19.00 · Applied Linear Algebra 18.10 · AI 18.14 · Formal Languages 18.19.

TECHNICAL SKILLS

Security: applied cryptography, SPA, network IDS sampling, adversarial examples for flow NIDS, pentest tooling, secure software development. Networking: SDN (Mininet, OVS), Wireshark, Scapy, Packet Tracer, incident-response process. Programming: Python, C/C++, Java, Kotlin, SQL, Bash, Assembly. Tools: Git, Linux, Docker, PyCryptodome, OpenSSL, Burp Suite, Metasploit, Nmap.

LANGUAGES

English: IELTS 7.5 (R 7.0, W 7.0, S 7.5, L 7.5), 4 December 2025. Persian: native.